



Introduction

Lemonity is a Unity Editor extension that allows you to navigate around the scene using your bare hands. A [Leap Motion](https://www.leapmotion.com/) device is required for hand tracking.

Check the latest version of this manual, including videos and animations at jonatanmartinez.com/lemonity

Requirements

- Leap Motion device in desktop mode
- One or more hands
- Windows 10 x64
- Scripting Runtime Version .NET 4.6 or above

Installation

- Lemonity requires the **Leap Motion Orion V4** service to be installed. You can download it for free here: <https://developer.leapmotion.com/setup/desktop>
- Import Lemonity package to your Unity project.
- Lemonity includes **LeapMotion Unity Core Assets 4.4.0**. If you already have this package added to your project, you should avoid importing it by unselecting “LeapMotion” folder. Alternatively, you can remove the folder from the project after importing (it will be in *Assets/Lemonity/LeapMotion*).

First Steps

1. Open Lemonity options window using Unity menu **Window > Lemonity Options**
2. Choose a **working mode** in the Lemonity Options window
3. Optionally open Lemonity Debug window: Window > Lemonity Debug to see more information about tracking.
4. Connect your Leap Motion device and put in on your desk.
5. You are ready to start working with Lemonity.

Gestures

In Lemony actions are triggered using two gestures:

Grab Gesture

Close all the fingers to make the grab gesture.



Pinch Gesture

Join together thumb and index fingertips. All the remaining fingers should stay opened.



Gesture limits can be adjusted in the [Options dialog](#).

Grab Mode

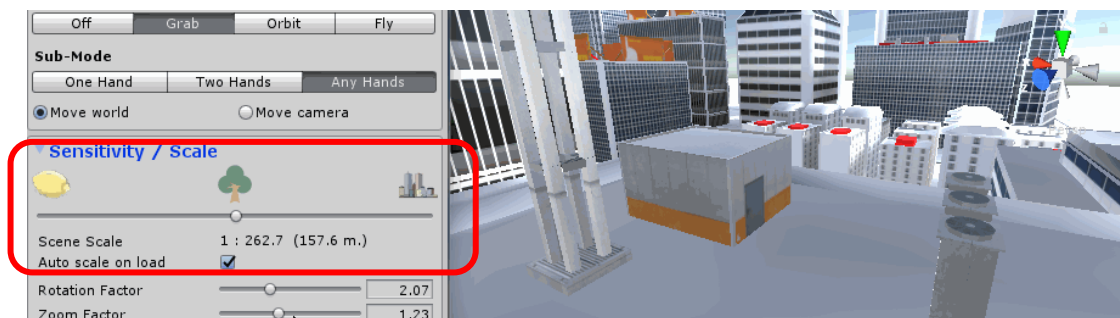
This working mode allows you to move and rotate the scene using direct manipulation with the **Grab Gesture**. You can use one or two hands.

As you move your closed hand, the scene moves according to the [Scene Scale](#) setting. You can also rotate the scene by rotating your hand. The speed of rotation is controlled by the [Rotation Factor](#).

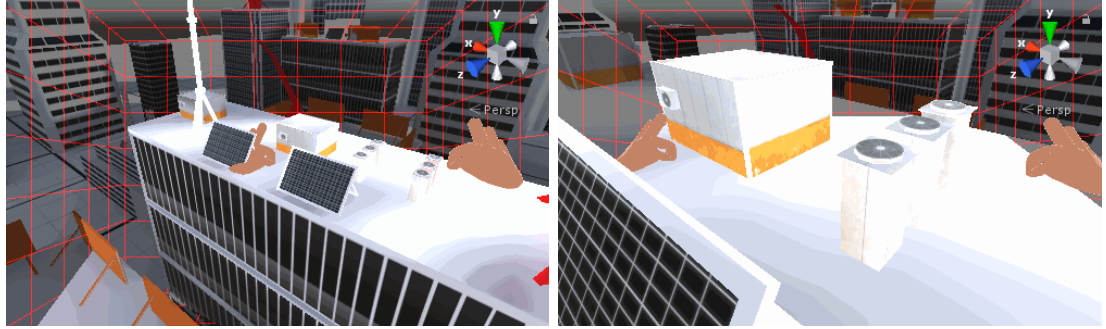
Setting the Scene Scale

The range of motion of your hands depends on the **Scene Scale** parameter. As the scale increases, the sensitivity and precision are decreased. It's convenient to enable **Show Workspace** (Visual Effects) to have an idea of the current range of motion.

- **Option 1.** Change the scale using the UI.



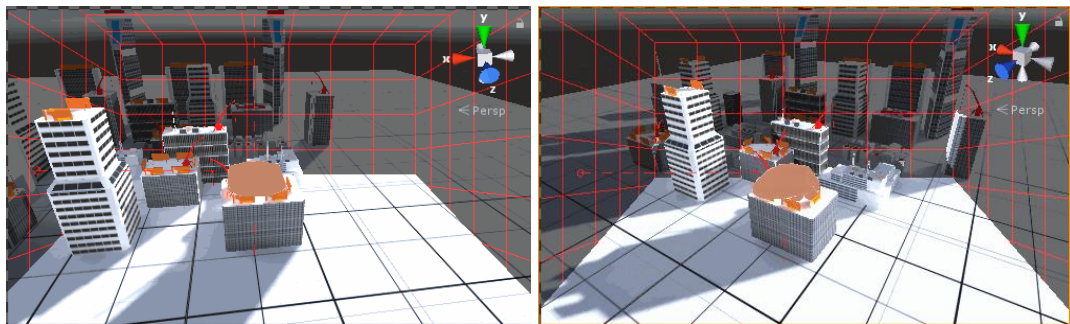
- **Option 2.** Change the scale using Pinch Gesture with both hands. Make the pinch gesture with both hands and move them apart as if you were zooming a picture in your phone.



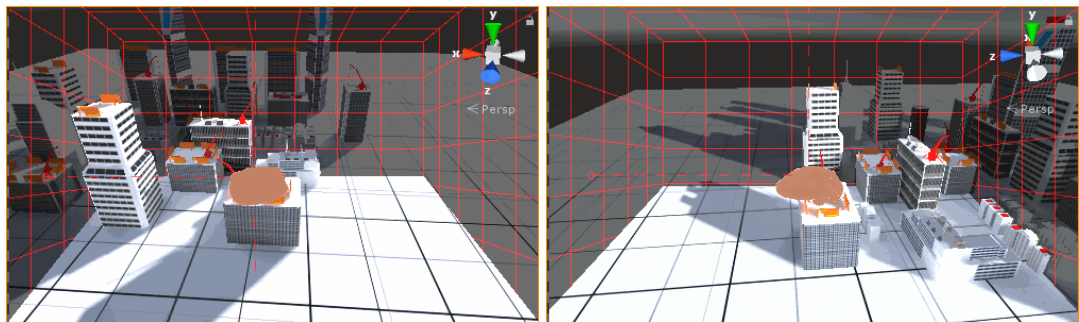
Rotation Factor

The rotation factor is a multiplier of the hand rotation. It allows you to perform rotations with less effort.

- Rotation factor **1x**



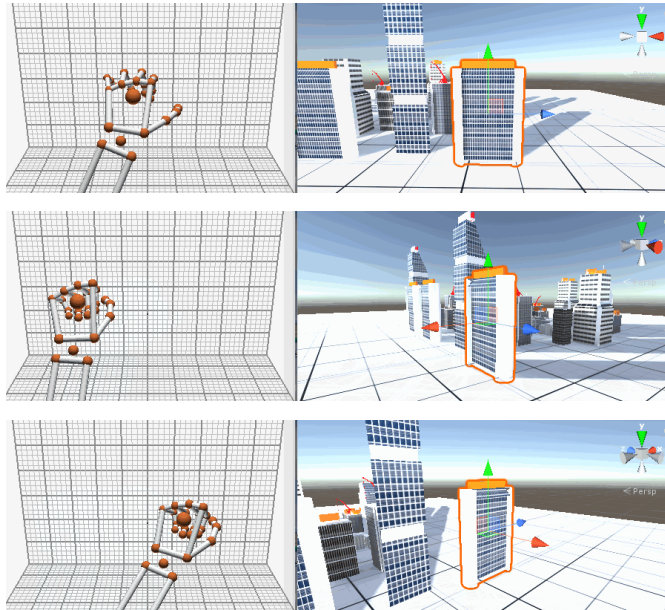
- Rotation factor **2x**



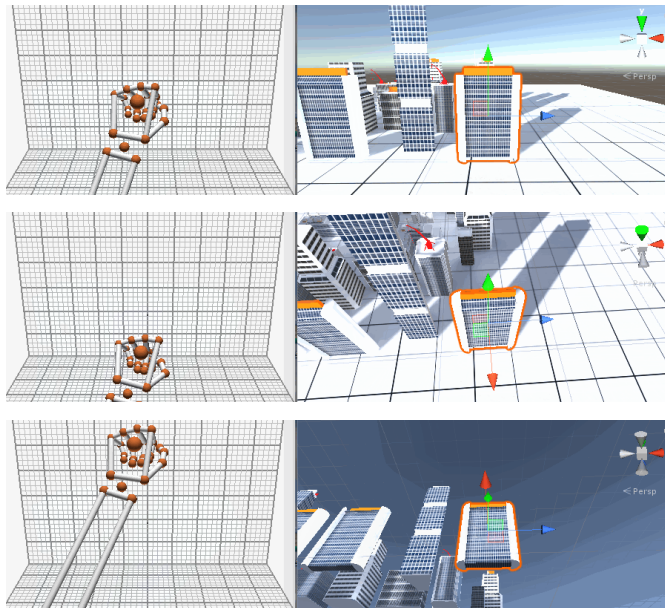
Orbit Mode

This mode allows you to orbit the camera around the selected object. The **grab gesture** controls the orbit:

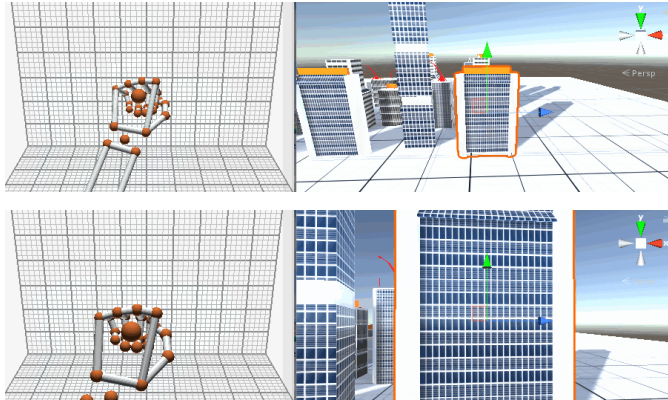
- Left/Right: Rotate around Y axis (yaw)



- Top/Bottom: Rotate around X axis (pitch)



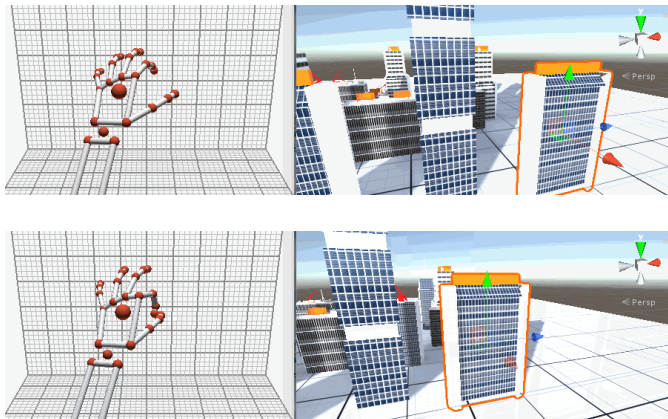
- Forward/Backward: Zoom



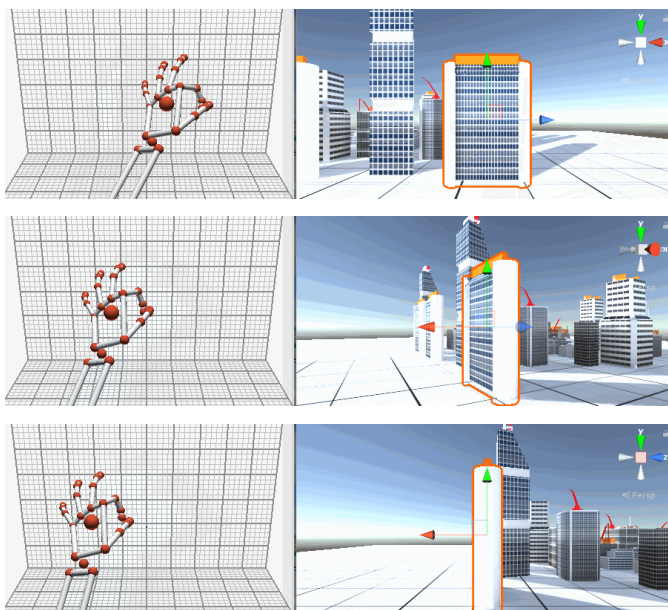
Align feature

Pinch gesture aligns the camera to the selected object:

- Simple pinch: Center view to selected object



- Pinch and move following an orthogonal axis: align view to axis

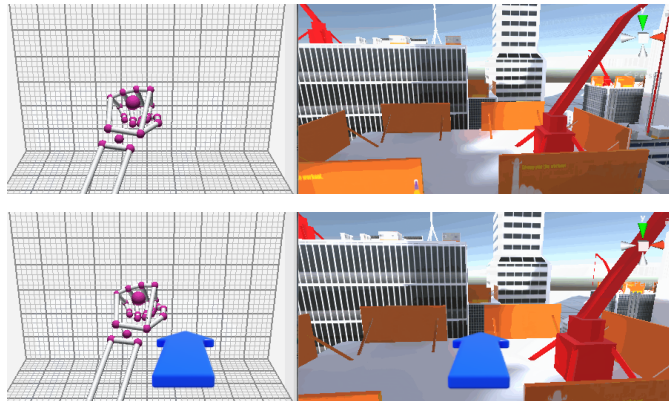


Fly Mode

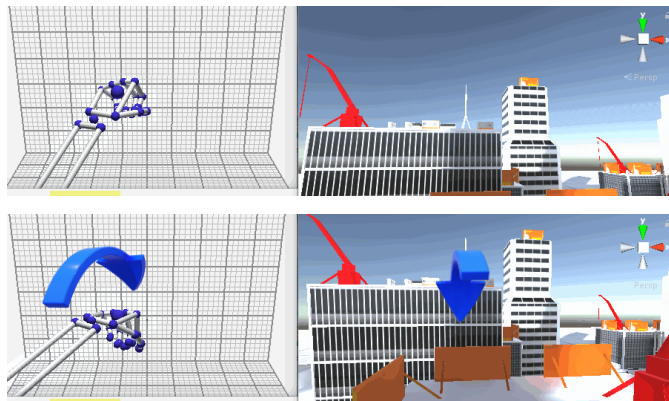
Fly around the scene using one or two hands using the **grab gesture**.

One hand

Move

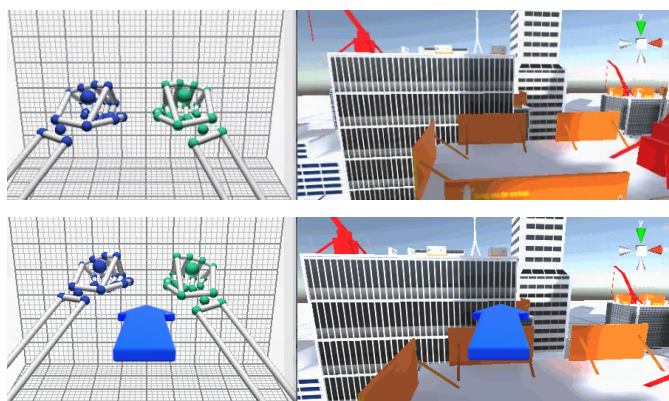


Rotate

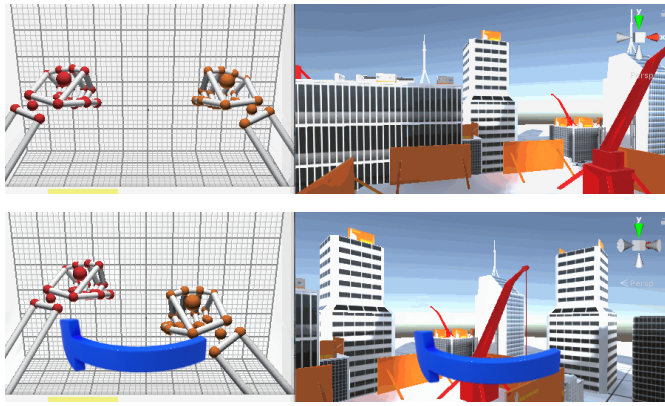


Two hands

Move

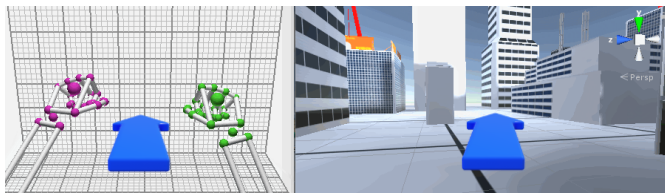


Rotate



Hover Mode

Hover mode is a special flight mode where the distance to the ground is fixed. We use the same gestures as in Fly mode



Options

Adjusting Gesture Thresholds

Gestures are triggered using hysteresis to improve stability. This means that they use a different threshold for detecting when a gesture starts and when it stops.

Grab gesture is detected using Leap Motion's *GrabStrength* parameter, which ranges from 0 when the hand is fully open, and 1 when all the fingers are closed.

Start Threshold: If *GrabStrength* is higher than this parameter then the gesture is triggered.

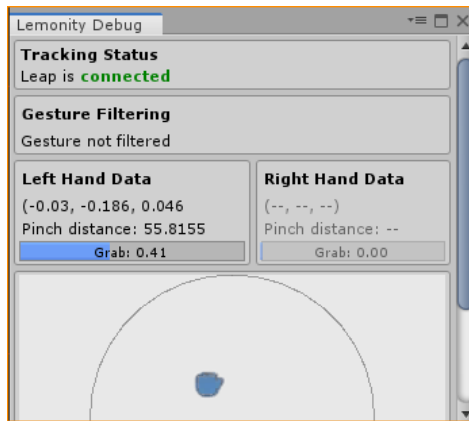
Stop Threshold: If *GrabStrength* is lower than this parameter then the gesture is released.

Pinch gesture is detected using the distance (in millimetres) from Index and Thumb fingertips. It ranges from a close value near zero when both fingers are touching, to more than 50 when the hand is open.

Start Distance: If the distance is lower than this parameter then the gesture is triggered.

Stop Distance: If the distance is higher than this parameter then the gesture is released.

It is recommended to open the Lemonity Debug window using the Unity menu Window > Lemonity Debug to check the recognition of gestures and fine tune the thresholds.



Tracking Filtering

Lemonity implements a [motion filter](#) to smooth the noise from the tracking. It filters Rotation and Position of the hands in space. The more aggressive the filter is, the smoother the movements will be at the cost of slow response.

Minimum Cut Off Frequency: Decrease it to decrease slow speed jitter

Beta Speed Coefficient: Increase the speed coefficient to decrease speed lag.

Cut Off for derivative: Used to smooth the speed calculation, defaults to 1.

Involuntary Gestures

Sometimes it is possible that a gesture is triggered involuntarily, especially when one of your hands moves near the Leap Motion workspace. This heuristic ignores the gestures that:

- Start outside of a defined safe zone (adjusted with the parameter **Safe Zone Radius**). This zone is showed in the Lemonity Debug Window.
- Move in outward direction.

The Lemonity Debug Window lets you know if a gesture is being filtered.

Frequently Asked Questions (FAQ)

Q: After adding the Lemonity package it doesn't work.

A: Probably Unity hasn't loaded the Leap Motion DLL yet. Please restart Unity.

Q: I've restarted Unity and it still doesn't work.

A:

- Open Lemonity Options Window (Alt+L or go to menu *Window/Lemonity Options*) and select a working mode (Grab/Orbit/Fly).
- Open Lemonity Debug Window (menu *Window/Lemonity Debug*) and check that Leap Motion is connected and that hands are detected if moved above Leap Motion.
- Open Leap Motion control panel and check it is working using their *Diagnostic Visualizer* (Troubleshooting tab)

Q: Lemonity does not remember the options I change.

A: Click the **Save** button on the bottom of the Lemonity Options window after you find a configuration you like.

Q: Gestures are not reliable enough. How can I improve them?

A:

- Open Lemonity Options Window and adjust thresholds in section **Gesture Thresholds**. You can test and fine tune them using the Lemonity Debug Window.
- Check that your gestures are not being ignored by the Involuntary Gestures Heuristic (see [Involuntary Gestures](#)). Increase the safe zone radius or disable the heuristic.

Q: Sometimes a gesture is triggered involuntarily.

A: Enable the Involuntary Gestures heuristic and/or decrease the safe zone radius (see [Involuntary Gestures](#)).

More Information

Official website: jonatanmartinez.com/lemonity

Do you need support or want to make any suggestion? Please write an email to: support@jonatanmartinez.com